

Paragraph Proofs and Flowchart Proofs (Homework)

Write a paragraph proof and a flowchart proof for each statement below.

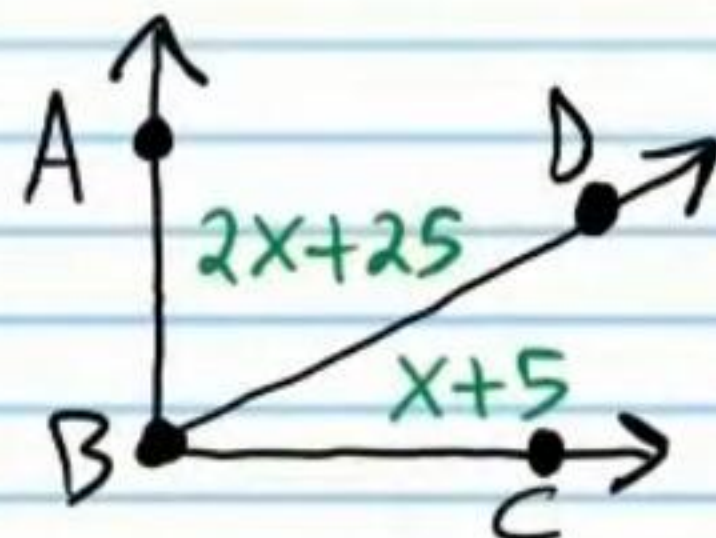
① If $\overline{AB} \cong \overline{CD}$ below, then $AB = 12$.



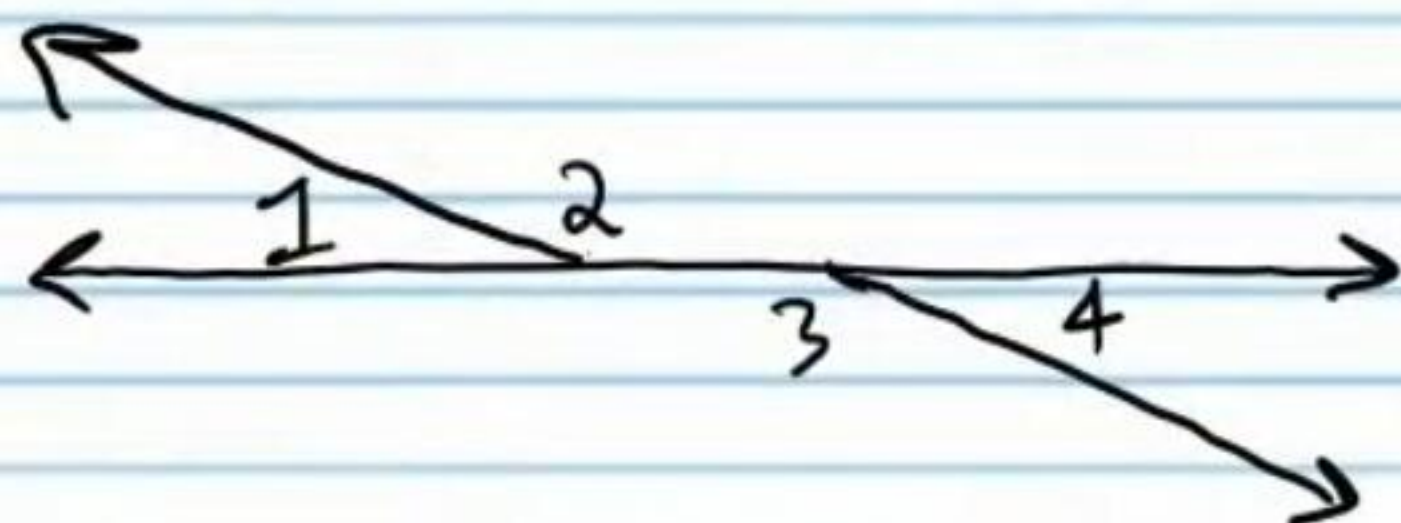
② If M is the midpoint of the segment below, then $AB = 2MB$.



③ If $\angle ABC$ below is a right angle, then $m\angle ABD = 65^\circ$.



④ If $\angle 2$ and $\angle 4$ below are supplementary, then $m\angle 2 = m\angle 3$. Use the Congruent Supplements Theorem (Covered in one of the early classes on proofs).

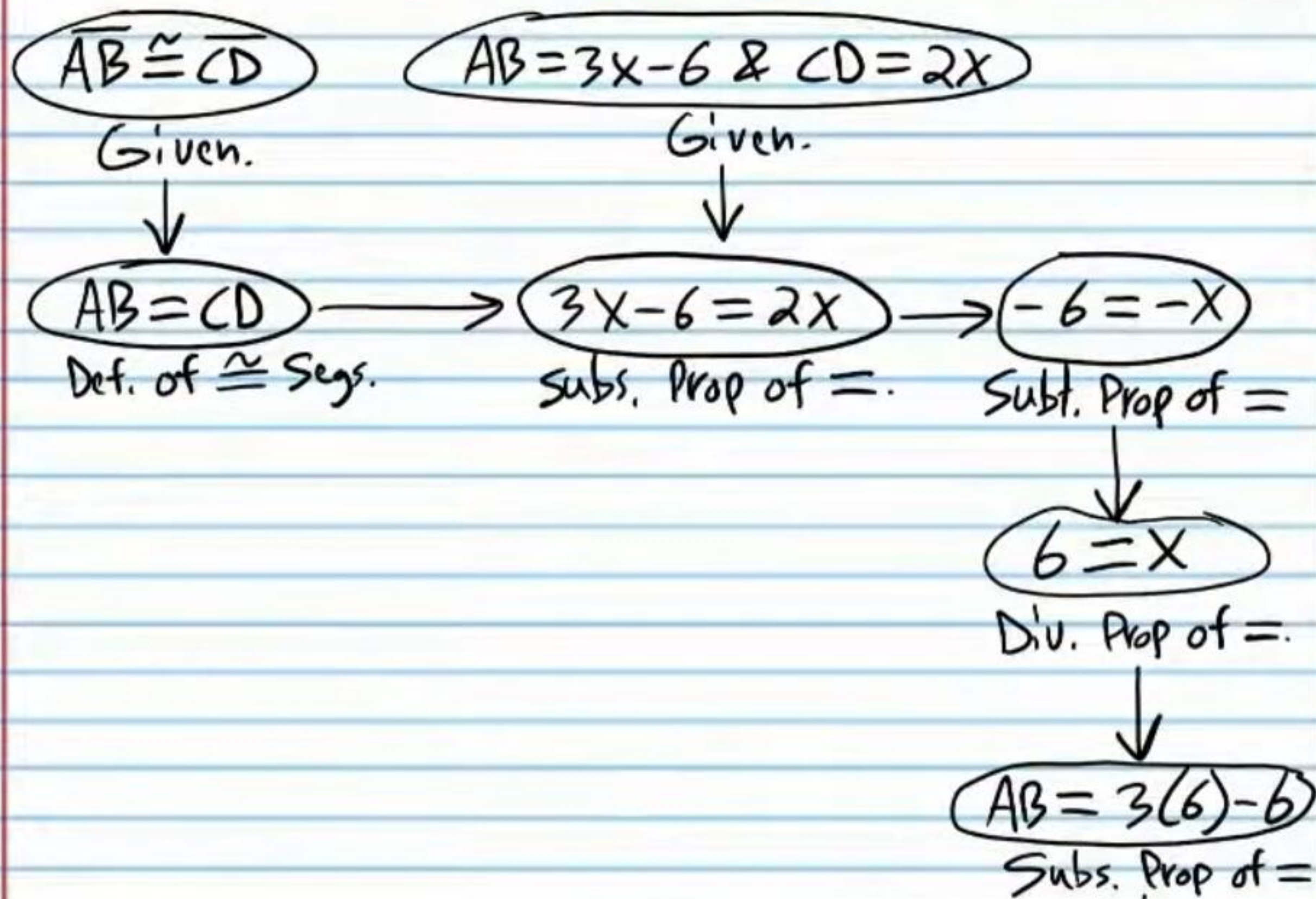


Answers

① Paragraph Form

It is given that $\overline{AB} \cong \overline{CD}$ and so $AB=CD$ by definition. The diagram shows that $AB=3x-6$ and $CD=2x$ and so $3x-6=2x$ by the Substitution Property of Equality. $-6=-x$ by the Subtraction Property of Equality and $6=x$ by the Division Property of Equality. $AB=3(6)-6$ by the Substitution Property of Equality and if this expression is simplified, then $AB=12$.

Flowchart Form



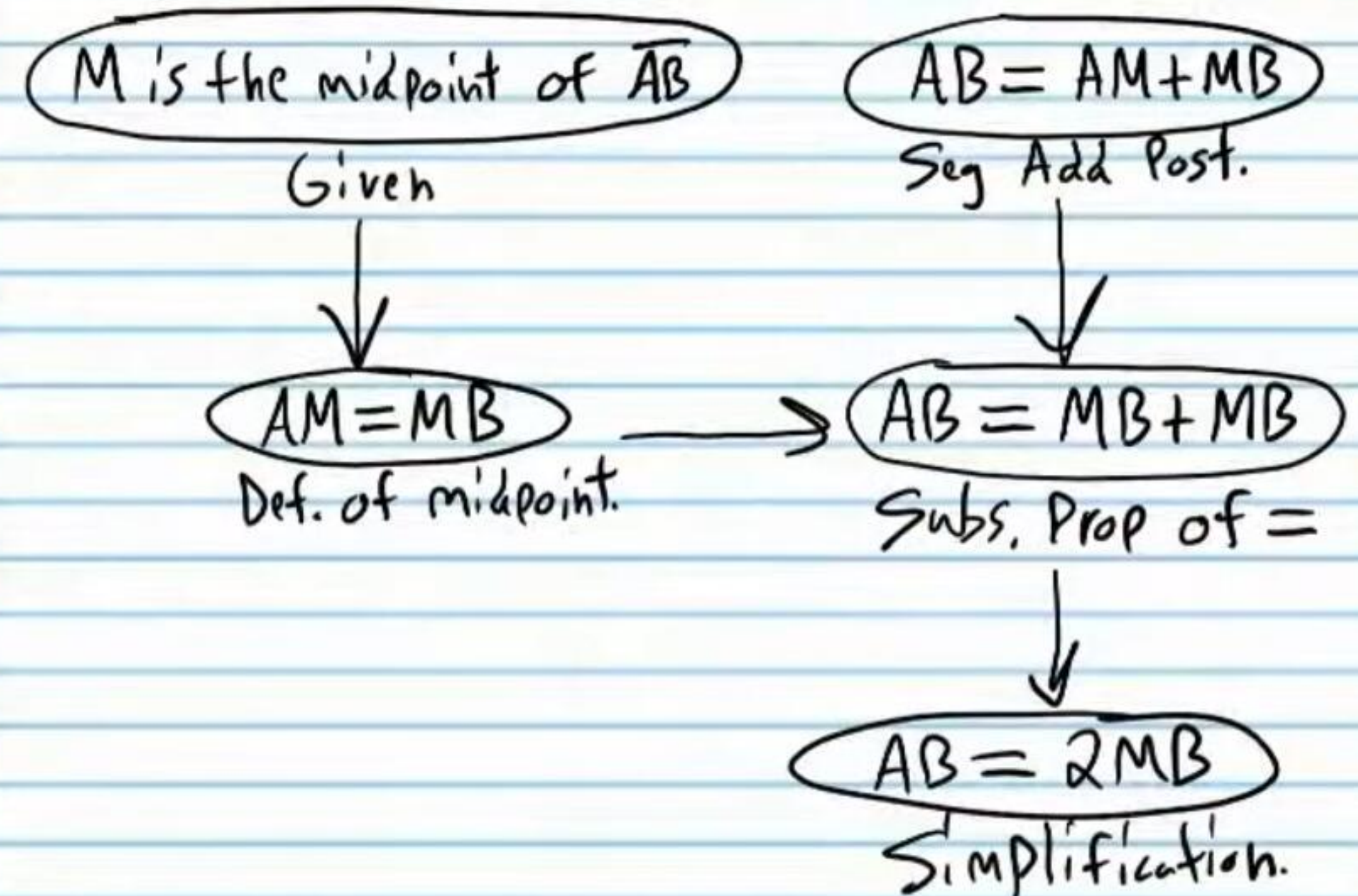
↓

$AB=12$
Simplification

② Paragraph Form

It is given the M is the midpoint of $\overline{AB}$ and so by the definition of midpoint, $AM=MB$. Furthermore, the Segment Addition Postulate states that $AB=AM+MB$ and by the Substitution Property of Equality, $AB=MB+MB$. Finally, this expression simplifies to $AB=2MB$.

Flowchart Form



③ Paragraph Form

By the Angle Addition Postulate, $m\angle ABC = m\angle ABD + m\angle DBC$ and the diagram shows that $m\angle ABD = 2x + 25$ and $m\angle DBC = x + 5$. Therefore, by the Substitution Property of Equality, $m\angle ABC = 2x + 25 + x + 5$ and when simplified this equation becomes $m\angle ABC = 3x + 30$. It is given that $\angle ABC$ is a right angle and so $m\angle ABC = 90^\circ$ by the definition of right angles. By the Substitution Property of Equality, $3x + 30 = 90$ and by the Subtraction Property of Equality, $3x = 60$. Furthermore, $x = 20$ by the Division Property of Equality and $m\angle ABD = 2(20) + 25$ by the Substitution Property of Equality. Finally, $m\angle ABD = 65^\circ$ if we simplify this resulting equation.

Flowchart Form

$$m\angle ABC = m\angle ABD + m\angle DBC$$

$\angle$ Add. Post.

$$m\angle ABD = 2x + 25 \text{ \& } m\angle DBC = x + 5$$

Diagram

$$m\angle ABC = 2x + 25 + x + 5$$

Subs. Prop of =

$$m\angle ABC = 3x + 30$$

Simplification.

$\angle ABC$ is a right angle.

Given

$$m\angle ABC = 90^\circ$$

Def. of rt $\angle$.

$$3x + 30 = 90$$

Trans. Prop of =.

$$3x = 60$$

Subt. Prop of =.

$$x = 20$$

Div. Prop of =.

$$m\angle ABD = 2(20) + 25$$

Subs. Prop of =.

$$m\angle ABD = 65^\circ$$

Simplification.

④

Paragraph Form

The diagram shows that $\angle 3$ and $\angle 4$ form a linear pair, and so by the Linear Pair Postulate, $\angle 3$ and $\angle 4$ must be supplementary. It is given that $\angle 2$ and $\angle 4$ are supplementary and therefore by the Congruent Supplements Theorem, $\angle 2 \cong \angle 3$. Finally, by the definition of congruent angles $m\angle 2 = m\angle 3$.

Flowchart Form

